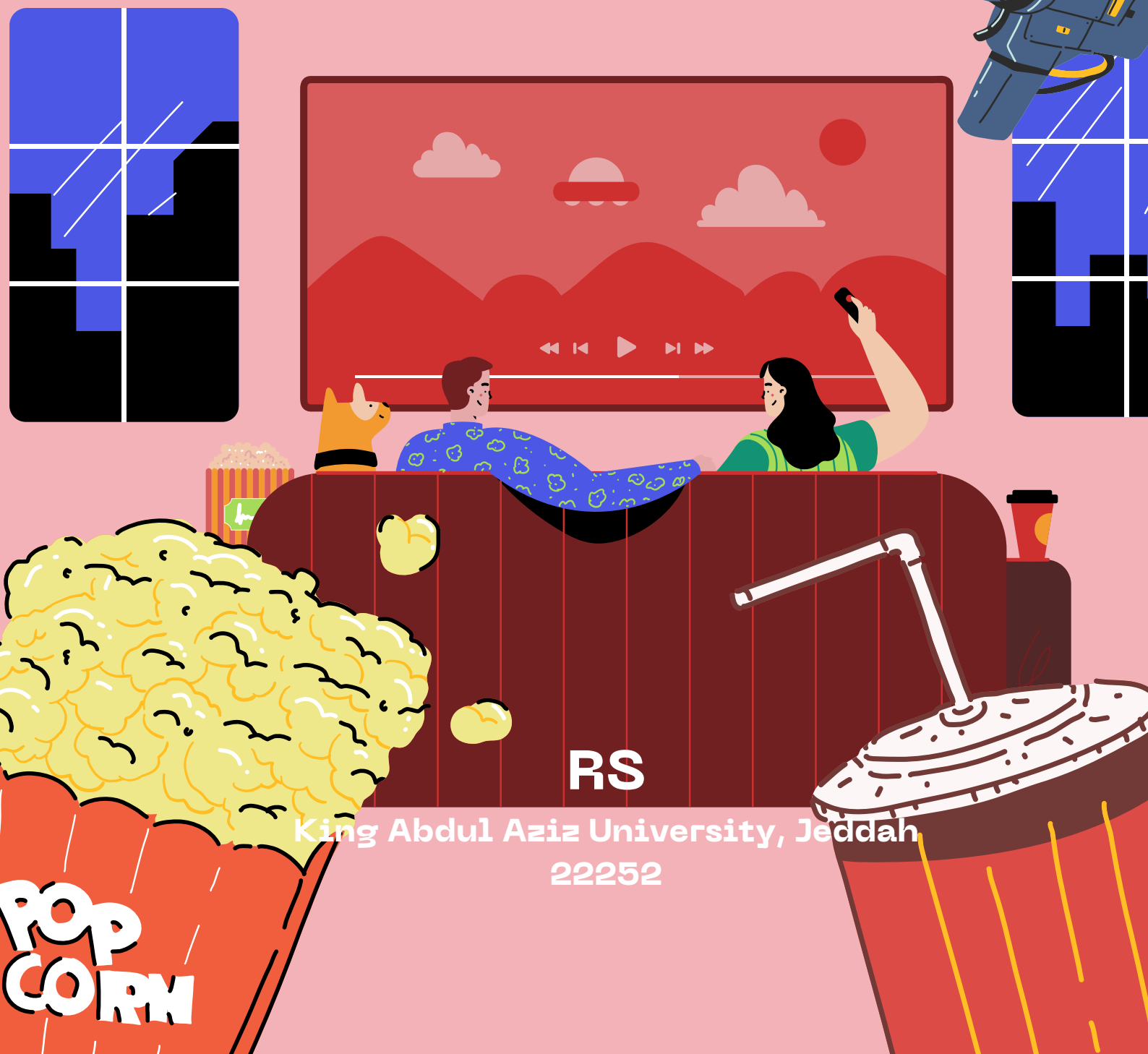
Mid Night

27 oct 2025 | 9:00 AM



RS

King Abdul Aziz University, Jeddah
22252

🔍 فهرس | Contents

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لولا الله ثم تعب وتعاون هذا الفريق لم يكن ليُكتب لهذا العمل الظهور بهذا الشكل

MED24 TEAM

أهيل الزهراني
جود الجهني
يزيد المجنوني
ندى النمر
عزيزه باداود
سلمان السنوسي
رهام صهلولي
غدى المالكي
غيداء الحربي
ميسان كلنتن
لينه السلمي

رؤوم الغامدي
نورا القصيمي
هاشم المكرمي
طلال النافعي
رواء حبيب الله
رؤى الوافي
ولاء النمري
نايف العتيبي
محمد المعيرفي
ارينا القرني
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ريم البلادي
إسراء أبوخماس
تركي حجي
ابوبكر الكثيري
أديب أبوشال
تاليا نوع
رتال البادي
معاذ الحارثي
محمد ظافر الشهري
عبدالرحمن باويان

لينه العطاس
محمد علي الشهري
سفانا الوذيناني
وجد الشهري
سارا الصاعدي
سامر المرغلاني
محمد شراحيلي
جواد العبيدي
ميس عامر
رنا السنوسي
ابراهيم الجهني
رزان احمد

شكراً لكم

محمد سندي

أماني الشمراني

بيان المالكي

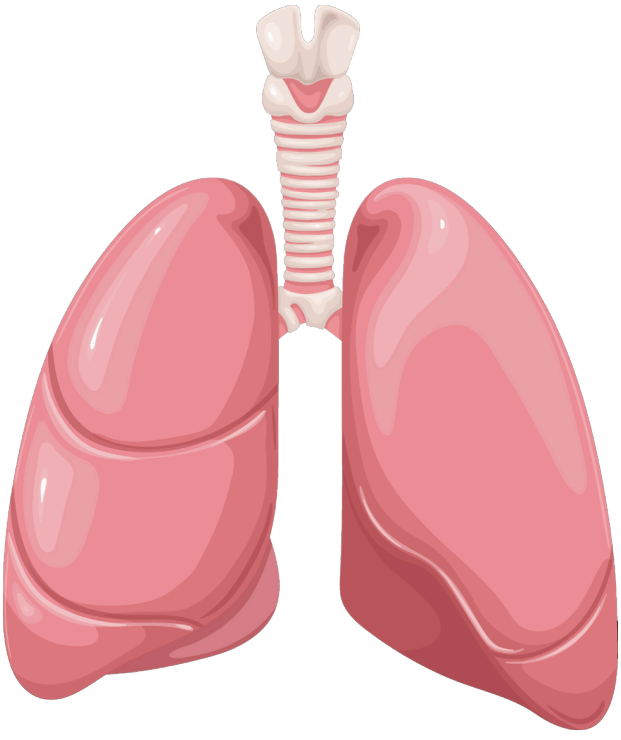
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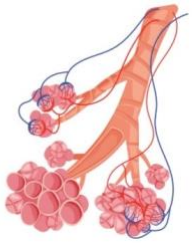
ليان الحربي

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اللهم إني أسألك فهم النبيين
وحفظ الملائكة المقربين،
اللهم اجعل لساني عامرًا بذكرك
وقلبي بخشيتك،
وسري بطاعتك،
إنك على كل شيء قدير
وحسبنا اللّٰه ونعم الوكيل



PHYSIOLOGY



L1 - Mechanism of respiration

(13 Q's)

1- Which one of the following is negative to atmospheric pressure?

A. Intra-alveolar Inspiration

B. Intra-alveolar Expiration

C. Intra-alveolar pressure equals zero

D. None of the above

Ans: A

Female slide 23

2- When is sternocleidomastoid used during breathing?

A. Deep expiration

B. At rest

C. Deep inspiration

D. At sleep

Ans: C

Female slide 26

3- During forced expiration what is the contracted muscle?

A. Sternocleidomastoid
B. Diaphragm
C. Abdominal muscles
D. External intercostal muscles

Ans: C

Female slide 26

4- Which of the following initiates the tidal expiration?

A. Elastic recoil of the lung
B. Intercostal muscle contraction
C. Contraction of the diaphragm
D. Forced blowing to excrete air from lung

Ans: A

Female slide 22

5- During expiration, the intra-alveolar pressure is?

A. Greater than atmospheric pressure
B. Lower than atmospheric pressure
C. Equal to atmospheric pressure
D. Equal to intrapleural pressure

Ans: A

Female slide 23

6- When you used the maximum adenosine triphosphate in respiration?

A. Tidal inspiration
B. Tidal expiration
C. Forced inspiration
D. Quite inspiration

Ans: C

Male slide 18

7- Which of the following statements is true?

A. During inspiration Intra-alveolar pressure is less than atmospheric pressure
B. Inspiration decrease venous return
C. In expiration Intra-alveolar pressure is less than atmospheric pressure
D. During inspiration intrpleural pressure is positive

Ans: A

Female slide 23

8- Which of the following is responsible for increased lymphatic return during inspiration?

A. Decreased transpulmonary pressure
B. Negative intraalveolar pressure
C. Abdominal muscle contraction

D. Increased negativity of intrathoracic pressure

Ans: D

Female slide 20

9- Contraction of which of the following muscles is responsible of the major lung compliance that increases lung volume ?

A. Sternocleidomastoide muscle

B. External intercostal muscles

C. Diaphragm

D. Internal intercostal muscles

Ans: C

Male slide 14

10- What is the primary force behind expiration during normal breathing cycle?

A. Abdominal muscles contraction

B. Elastic recoil

C. Increase intrapleural pressure

D. Transmural pressure gradient

Ans: A

Female slide 22

11- What is cellular respiration ?

A. Use of O2 by mitochondria to produce energy from food metabolism.

B. The transport of 02 in the blood circulation.

C. O2 and CO2 exchange in the tissue level.

D. O2 and CO2 exchange between air and lung.

Ans:A

Female slide 4

12- In a project representing lung ventilation, where the lung is depicted as an elastic material, the diaphragm as a dome shaped rubber, and the pleura as a balloon, what structure initiate the inspiration?

A. The elastic material

B. The rubber stretches..

C. The balloon.

D. None of the above.

Ans:B

Female slide 22

13- What is the primary force behind expiration during normal breathing cycle?

A. Abdominal muscles contraction

B. Elastic recoil

C. Increase intrapleural pressure

D. Transmural pressure gradient

Ans: A

Female slide 22

L2 - Pressure - Volume relationship

(14 Q's)

1- Lung compliance increased in?

A. Pulmonary Fibrosis

B. Pulmonary edema

C. Pulmonary emphysema

D. Surfactant deficiency

Ans: C

Female slide 17

2- Deficiency in pulmonary surfactant leads to?

A. Decrease surface tension

B. Decrease lung compliance

C. Decrease filtration in alveoli

D. Increase FEC

Ans: B

Female slide 10

3- During inspiration?

A. Expend less energy than expiration

B. Increase transmural pressure

C. Intra-pleural pressure increases

D. Venues return decrease

Ans: B

Female slide 6

4- Regarding surfactant, which of the following is true?

A. It reduces surface tension

B. It increases work of breathing

C. It increases alveolar collapse

D. It causes bronchoconstriction

Ans: A

Female slide 11

5- Which of the following increases lung compliance?

A. Old age
B. Emphysema
C. Lung fibrosis
D. Obesity

Ans: A&B

Female slide 17

6- What is the role of surfactant in breathing in a healthy subject?

A. Enhance lung immunity
B. Prevent alveolar collapse
C. Facilitate gas diffusion
D. Reduce lung expansion

Ans: B

Female slide 11

7- Which of the following statements is true regarding intrapleural pressure?

A. Increases during inspiration
B. Equal to the atmospheric pressure
C. Sub-atmospheric pressure
D. Equals to the alveolar pressure

Ans: C

Female slide 5

8- Which of the following prevent lung collapse during expiration?

A. Surfactant
B. Increase negativity of intrapleural pressure during inspiration
C. Increase negativity of intra alveolar pressure during inspiration
D. Transmural pressure

Ans: A

Female slide 11

9- What prevents fluid from moving from the pulmonary capillaries into the alveoli?

A. The repelling forces of the surfactant
B. Negative intrapleural pressure
C. Negative transthoracic pressure
D. Pulmonary Hydrostatic pressure

Ans: A

Female slide 11

10- Which of the following prevent leakage of fluid from blood capillaries to Alveoli?

A. Repull surfactant force

B. Negativity of intrathoracic pressure

C. Negativity of intrapleural pressure

D. Hydrostatic pressure

Ans: A

Female slide 11

11- What prevents fluid from leaking out of the capillaries into the alveoli?

A. Hydrostatic pressure

B. Surfactant

C. Intra-pulmonary pressure

D. Intra-thoracic pressure

Ans: B

Female slide 11

12- A 70-year-old man who has been a heavy smoker for nearly 50 years with possible harm to type II alveolar cells. Which of the following would probably decrease?

A. Lung compliance

B. Filtration into the alveoli

C. Work of breathing

D. Surface tension in the alveoli

Ans: A

Female slide 9

13- An 18 years old man presents to the emergency department after a stab wound to the chest. He is combative and with severe shortness of breath. What will happen to him?

A. Intrapleural pressure increased with possible lung collapse
B. No healing
C. Increase of intrathoracic pressure
D. Interalveolar pressure increased with possible lung collapse

Ans: A

Normally, the pressure in the pleural cavity is negative, keeping the lungs inflated. A penetrating chest injury allows air to enter the pleural space, increasing intrapleural pressure

14- A patient came in from a car accident and had a crush injury, he is presented with shallow rapid breathing, and chest pain .

What is the most likely situation?

A. Increased intrapleural pressure and right lung collapse
B. Increased intrapleural pressure and no effect on the right lung
C. Increased intra alveolar pressure
D. Overinflation of the lung

Ans: A The chest injury allows air to enter the pleural space, increasing intrapleural pressure leading to lung collapse

15- What prevents pulmonary edema ?

A. -ve Intrathoracic pressure.
B. +ve Intrapleural pressure.
C. -ve Intrapleural pressure.
D. Repelling force of surfactant.

Ans:D

Female slide 12

16- Which of the following is responsible for increased lymphatic return during inspiration?

A. Decreased transpulmonary pressure
B. Negative intraalveolar pressure
C. Abdominal muscle contraction
D. Increased negativity of intrathoracic pressure

Answer: D

Female slide 15

L3 - Gas Diffusion and blood flow

(11 Q's)

1- Which of the following, if reduced, will decrease the rate of diffusion of the gas?

A. Molecular size

B. Fibrosis

C. Surface area of alveoli

D. Thickness of alveolar membrane

Ans: C

Female slide 31

2- Fick's law of diffusion is regarding?

A. Diffusion and surface area

B. Diffusion and pore diameter

C. Diffusion and water pressure

D. All of the above

Ans: A

Female slide 29

3- Which of the following factors causes increased alveolar PCO2?

A. Increased respiratory membrane thickness

B. Increased oxygen concentration

C. Severe bronchoconstriction

D. Decreased lung compliance

Ans: C

Female slide 14

4- Compared to the base of the lung, in a person who is in a recumbent position, the apex of the lung has?

A. Higher ventilation rate

B. Higher perfusion rate

C. Higher ventilation/perfusion (V/Q) ratio

D. The same V/Q ratio

Ans: D

The ventilation and perfusion differences from the apex to the base of the lung have usually been attributed to gravity: they tend to disappear in the supine position (source: Ganong's)

5- What occurs as a result to hyperventilation?

A. Increased PO2 increased PCO2

B. Decreased PO2 increased PCO2

C. Increased PCO2 decreased PH

D. Decreased PCO2 increased PH

Ans: D

Female slide 15, Male slide 16

6- What is the pressure of CO2 in the lungs?

A. 40

B. 100

C. 90

D. 50

Ans: A

Female slide 24

7- Why arterial blood shows increased oxygen saturation than venous blood ?

A. Due to increased PaO2 in arterial blood more than venous blood

B. Due to increased PaCO2 in arterial more than venous blood

C. Due to decreased amount of hemoglobin in venous blood

D. Due to increased temperature in arterial blood increase oxygen saturation of hemoglobin

Ans: A

Female slide 24

8- What happens to a patient who smokes that makes gas diffusion more difficult?

A. Iron deficiency anemia

B. Increasing thickness of respiratory membrane

C. Increase red blood cell

D. Increase surface area

Ans: B

Female slide 32

9- Hypothetically if drug X blocks the alveolar cells ATP synthesis what will happen to the diffusion rate?

A. It will affect the diffusion on CO2 rather than O2

B. No effect on the diffusion rate

C. Decrease the diffusion rate

D. Increase the diffusion rate

Ans: B

Female slide 32&33

10- Someone has an obstruction in the bronchus, what is the expected PO2 in the capillaries?

A. Equal to the partial pressure of atmospheric

B. Equal to normal arterial PO2

C. More than PO2 of inspired air

D. Equal to venous PO2

Ans: D

Not mentioned directly in slides

11- A woman has an abdominal surgery and 2 weeks later comes back with shortness of breath and chest pain , and her left leg is swollen with the overlying skin being a reddish blue color. What is the physiological compensatory response?

A. Contraction of local pulmonary arterioles
B. Relaxation of local arteriole smooth muscle
C. Contraction of bronchioles smooth muscle
D. Decrease in airway resistance

Ans:C

Female slide 12

L4- Physiology & transport of oxygen

(8 Q's)

1- Why is normal adult hemoglobin considered an efficient O2 carrier?

A. Hb has changeable affinity to O2 according to bioavailability
B. Due to highest affinity of binding to O2 compared to other gasses
C. Due to linear relationship between PaO2 and % hemoglobin saturation D. Due to the stability of Hb-O2 binding

Ans: A

Female slide 4

2- Carboxyhemoglobin in the blood indicates that hemoglobin (Hb) is combined with?

A. Carbon dioxide
B. Oxygen
C. Carbon monoxide
D. Nitrogen

Ans: C

Female slide 16

3- Regarding tissue hypoxia, which of the following is true?

A. Left to right shunt can cause hypoxia
B. Occur only when oxygen saturation low
C. Carbon monoxide can cause hypoxia
D. Is unlikely to happen in Polycythemia

Ans: C

Female slide 16

4- What is the most common form of oxygen carried in the blood?

A. Physically dissolved in plasma
B. Bound to hemoglobin

C. Bound to plasma protein

D. As bicarbonate

Ans: B

Female slide 4

5- Increased acidity (lower pH) in the blood is likely to have which of the following effects?

A. Decreased oxygen delivery to the tissue

B. Increased affinity of HB to oxygen

C. Increased oxygen delivery to the tissue

D. Decreased affinity of HB to CO2

Ans: C

Female Slide 14

6- Why is hemoglobin the suitable carrier of 02 in human blood?

A. Greater affinity for O2 than for CO

B. Curve shift to the left as a result of alkaline pH

C. Releases most of its content when PO2 between 5 and 20 mmHg

D. O2 binds loosely and reversibly with hemoglobin

Ans: D

Female slide 5

7- What makes hemoglobin an effective carrier of oxygen in the body?

A. It binds reversibly to oxygen

B. It binds irreversibly to oxygen

C. It dissolves a small amount of oxygen in plasma

D. It requires ATP to bind oxygen

Ans: A

Female slide 6

8- The increase in CO_2 and acidity that lowers the affinity of hemoglobin for oxygen is known as:

A. Bohr effect

B. Haldane effect

C. Hering–Breuer reflex

D. Dalton’s law

Ans: A

Female slide 23

L5- Carbon Dioxide Transportation

(7 Q's)

1- Most CO_2 is transported in blood as?

A. Carbamino/hemoglobin

B. Bound to reduced hemoglobin

C. Dissolved in plasma

D. Bicarbonate

Ans:D

Female slide 5

2-Which of the following statements about CO_2 levels in the alveoli is correct?

A. The partial pressure of oxygen affects CO_2 levels

B. CO_2 levels in the alveoli remain constant

C. Ventilation does not influence CO_2 levels in the alveoli

D. CO_2 diffuses from the alveoli into red blood cells in the bloodstream

Ans: A

Female slide 6&7

3- What is the percentage of carbon dioxide in plasma?

A. 80%

B. 40 %

C. 10%

D. 20 %

Ans: C

Female slide 5

4- CO_2 is mainly transported in the blood in the form of?

A. Dissolved in plasma

B. Bind to proteins

C. Carboxy-Hb

D. Bicarbonate

Ans: D

Female slide 5

5- The concentration of CO_2 is highest in which of the following?

A. Bicarbonate

B. Dissolved in plasma

C. Carbamino acid

D. Oxygenated hemoglobin

Ans: A

Female Slide 5

6- Which of the following statements is true about CO2?

A. It's more soluble in blood than oxygen
B. Its dissociation curve is sigmoidal in shape
C. Tidal CO2 adds 20 cc to arterial blood
D. None of the above

Ans: A

Female slide 5

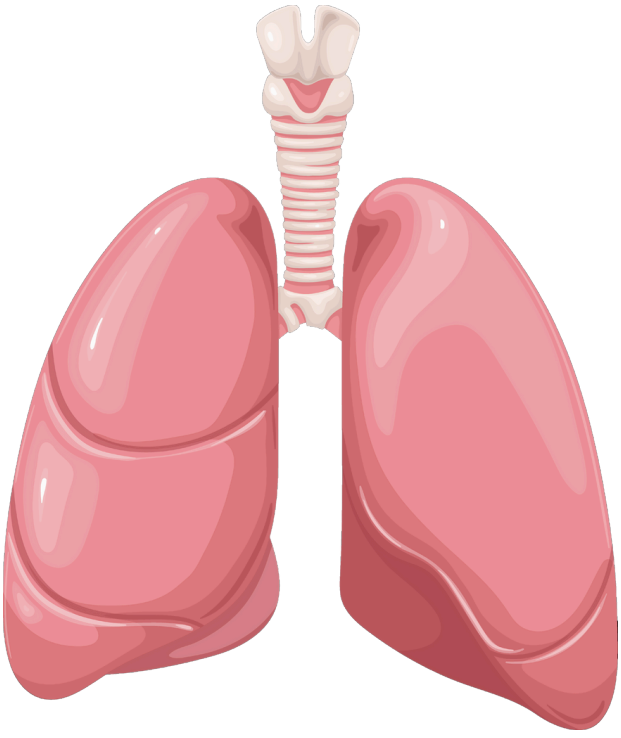
7- CO_2 is mainly transported in the blood in the form of:

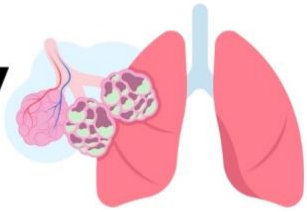
A. Dissolved in plasma
B. Bind to proteins
C. Carboxyhemoglobin
D. Bicarbonate

Ans: D

Female slide 2

اللهم يا جامع النّاس في يوم لا ريب فيه
اجمع عليّ ضالتي



PATHOLOGY 

L1- Atelectasis

(25 Q's)

1- Bronchiectasis causes?

A. Resorption atelectasis

B. Compression atelectasis

C. Contraction atelectasis

D. Restrictive airway disease

Ans: A

Female slide 8

2- Which of the following diseases cause a decreased FEV1/FVC ratio?

A. Chronic Bronchitis

B. Pneumoconiosis

C. Acute interstitial lung disease

D. Sarcoidosis

Ans: A

Female slide 8

3- A patient is in a car accident and suffers major trauma in the mediastinum and chest wall that leads to hemothorax. Many major arteries are damaged and the patient is unable to catch his breath. What is the cause of this?

A. Resorption Atelectasis
B. Contraction Atelectasis
C. Compression Atelectasis
D. Restrictive lung disease

Ans: C

Female slide 9

4- (Case) A child came into the ER, after aspiring a strawberry that has lodged into his right bronchi, what is the likely result?

A. Contraction atelectasis
B. Constrictive atelectasis
C. Resorption atelectasis
D. Pneumothorax

Ans: C

Female slide 8

5- (Case) A patient came in with interstitial fibrosis. What may be the result of the pulmonary function test?

A. FVC reduced
B. TLC normal
C. FEV1/FVC decreased

D. TLC increased

Ans: A

Female slide 15

6- (Case) A man was stabbed in the right 5th intercostal space at the midaxillary level. X-ray showed air in the pleura. What's the most probable complication of the underlying case?

A. Atelectasis

B. Pneumonia

C. Congestive heart failure

D. Pneumoconiosis

Ans: A

Female slide 8

7- (Case) Hemothorax causes?

A. Contraction atelectasis

B. Compression atelectasis

C. Resorption atelectasis

D. Pneumothorax

Ans: B

Female slide 9

8- A 6 year old child aspirates a peanut and comes into the ER with shortness of breath, dyspnea, and light-headedness. What is the cause of his symptoms?

A. Compression Atelectasis
B. Contraction Atelectasis
C. Resorption Atelectasis
D. Pneumothorax

Ans: C

Female slide 8

9- A patient is in a car accident and suffers major trauma in the mediastinum and chest wall. Many major arteries are damaged and the patient is unable to catch his breath. What is the cause of this?

A. Resorption Atelectasis
B. Contraction Atelectasis
C. Compression Atelectasis
D. Pneumoconiosis

Ans: C

Female slide 9

10- A middle age men present to clinic with chronic bronchitis. After the spirometry result show ?

A. lncreased FEV1/ FVC AND increased TLC

B. Decreased FEV1/ FVC AND decreased TLC

C. Decreased FEV1/ FVC AND increased TLC

D. Increased FEV1/ FVC AND normal TLC

Ans: C

Female slide 17

11- Resorption atelectasis is mainly caused by?

A. Lung fibrosis or pleura

B. Pleural effusion

C. Commonly in bedding patients

D. Obstructive bronchi

Ans: D

Female slide 8

12- Bronchiectasis causes?

A. Contraction atelectasis

B. Resorption atelectasis

C. Reduced forced vital capacity

D. Compression atelectasis

Ans: B

Female slide 8

13- Atelectasis cause hypoxia by which of the following?

A. Ventilation perfusion imbalance

B. Not cause hypoxia

C. Fused but not collapse

D. Ventilation perfusion balance

Ans: A

Female slide 4

14- Bronchiectasis causes which of the following complications?

A. Contraction atelectasis

B. Compressive atelectasis

C. Resorption atelectasis

D. Restrictive lung disease

Ans: C

Female slide 8

15- The disease that reduced the FEV1/FVC ratio?

A. Pneumoconiosis

B. Sarcoidosis

C. Chronic bronchitis

D. Idiopathic pulmonary fibrosis

Ans: C

Female slide 8

16- A 6 year old child aspirates a peanut and comes into the ER with shortness of breath. After 3 days he had a fever and diarrhea. What is the complication that may happen to him?

A. Bronchopneumonia
B. Bronchiectasis
C. Resorption
D. Acute bronchitis

Ans: C

Female slide 7

17- (Case) A smoker came to the clinic with dyspnea and chronic productive cough that lasted 3 months over the last 2 years. What is true about the laboratory findings?

A. Reduced FEV1/FVC ratio
B. Normal FEV1/FVC ratio
C. Decreased total lung capacity
D. Reduced FVC

Ans: A

Female slide 16

18- What is correct about acute respiratory distress syndrome?

A. It is a restrictive lung disease
B. FVC is increased

C. TLC increased

D. FEV1/FVC decrease

Ans: A

Female slide 17

19- Which of the following will most likely result in contraction atelectasis?

A. Pleural fibrosis

B. Pneumothorax

C. Bronchial asthma

D. Bronchiectasis

Ans: A

Female slide 10

20- The hallmark of radiological appearance of lung collapse on chest radiograph?

A. Movement of normal anatomical structure

B. Air Bronchogram sign

C. Heterogeneous opacity with no significant mass effect

D. Contralateral deviation of the mediastinum

Ans: A

Female slide 6

21- In Lung collapse, which of the following is noticed?

A. Volume increases
B. Movement of anatomical structures
C. Air Broncho gram sign
D. Contralateral deviation of the mediastinum

Ans: B

Female slide 6

22- A smoker came to the clinic with dyspnea and chronic productive cough that lasted 3 months over the last 2 years. What is true about the laboratory findings?

A. reduced FEV1/FVC ratio
B. Normal FEV1/FVC ratio
C. Decreased total lung capacity
D. DI Reduced FVC

Ans:A

23- A 60-year-old male with a history of heavy smoking presents with persistent cough and shortness of breath for 2 months. Chest X-ray shows a right upper lobe mass, and the surrounding lung appears collapsed. The patient denies fever or recent respiratory infections

A. Resorption Atelectasis
B. Compression Atelectasis
C. Constrictive Atelectasis

Ans:A

24-A smoker came to the clinic with dyspnea and chronic productive cough that lasted 3 months over the last 2 years. What is true about the laboratory findings?

A. Reduced FEV1/FVC ratio

B. Normal FEV1/FVC ratio

C. Decreased total lung capacity

D. Reduced FVC

Ans: A

Female slide 15

25-A 60-year-old male with a history of heavy smoking presents with persistent cough and shortness of breath for 2 months. Chest X-ray shows a right upper lobe mass, and the surrounding lung appears collapsed. The patient ?denies fever or recent respiratory infections. What is the type of atelectasis

A. Resorption Atelectasis

B. Compression Atelectasis

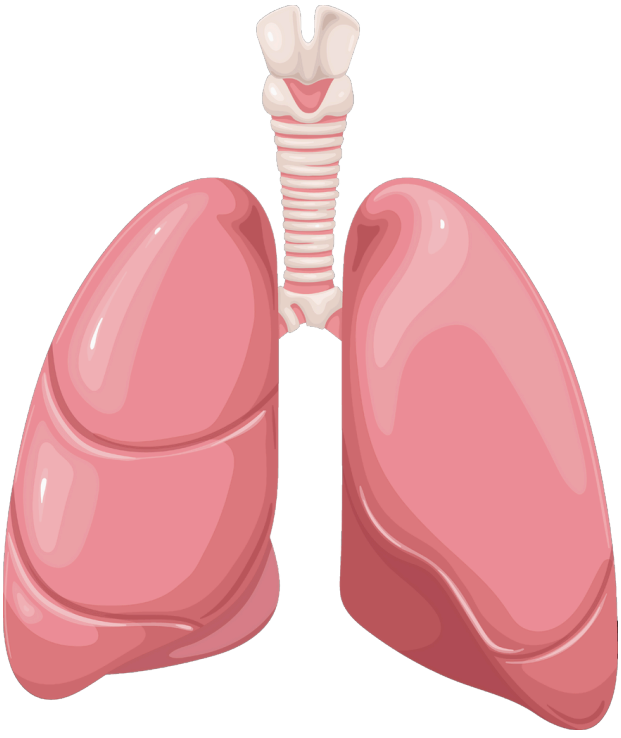
C. Constrictive Atelectasis

D. Patchy Atelectasis

Ans: A

Patho slide 8

اللهم ذكرني ما نسيت و علمني ما جهلت



HISTOLOGY 

L1- Histology of the respiratory system

(14 Q's)

1- Which of the following is a feature of the trachea?

A. Submucosa contains numerous seromucus glands

B. C-shaped fibrocartilage

C. Submucosacontainssmoothmuscles

D. None

Ans: A

Female slide 10

2- Regarding the lung, which of the following statements is correct?

A. The intrapulmonary bronchi are deficient of cartilage

B. Alveolar sacs and ducts are parts of the conducting portions of respiratory system

C. Therespiratorybronchiolesarelinedbypseudo stratified columnar ciliated epithelium

D. The alveoli intercommunicate through alveolar pores

Ans: D

Female slide 17

3- Which of the following is CORRECT regarding the terminal bronchiole?

A. They contain Clara cells
B. They have alveoli on their walls
C. They are lined with stratified squamous epithelium
D. They are the site of gas exchange

Ans: A

Female slide 15

4- All of the following statements about the respiratory epithelium are true, except:

A. The respiratory epithelium has Type I pneumocytes which secrete surfactant
B. The respiratory epithelium has cilia
C. The respiratory epithelium covers some of the conducting parts of the respiratory system
D. The respiratory epithelium plays a role in filtering and humidifying inhaled air

Ans: A

Female slide 4 & 17

5- The bronchioles differ from the bronchi in all of the following, EXCEPT?

A. Absence of cartilage

B. Presence of smooth muscles and elastic fibers

C. Epitheliumtype

D. Presence of Clara cells

Ans: B

Female slide 14

6- Which of the following cells function as a blood-air barrier?

A. Pneumocytes I

B. PneumocytesII

C. Gobletcell

D. Clara cell

Ans: A

Female slide 19

7- What are the changes in respiratory epithelium occur during the passage from the trachea to the terminal bronchioles?

A. Simple squamous epithelium changes to pseudostratified columnar ciliated epithelium
B. Simple cuboidal partially ciliated epithelium changes to simple columnar epithelium
C. Pseudostratified columnar ciliated epithelium changes to ciliated simple cuboidal epithelium
D. Simple columnar ciliated epithelium changes to squamous epithelium

Ans: C

Female slide 15

8- What is the function of respiratory epithelium?

A. Transport of the oxygen from capillary to alveoli
B. Trapping the air particles and clear it
C. warmingtheair
D. Immune Response

Ans: B

Female slide 4

9- Which of the following have ciliated simple columnar epithelium and are devoid of cartilage and glands?

A. Alveolar ducts

B. Bronchioles

C. Secondarybronchus

D. External bronchi

Ans: B

Female slide 14

10- Which of the following does not contain smooth muscle?

A. Alveoli

B. Trachea

C. Bronchi

D. Bronchioles

Ans: A

Female slide 17, 18 , 19

11- The capillaries in inter-alveolar septa are?

A. Continuous non-fenestrated

B. Discontinuousfenestrated

C. Separated from air only by its own epithelium and lamina

D. Fenestrated to allow the air from the alveoli

Ans: A

Female slide 19

12- Intrapulmonary bronchi differs from extra pulmonary bronchi in which of the following?

A. Type of epithelium

B. Contain shared cartilage

C. Have muscularis smooth muscle

D. Presence of submucosal glands

Ans: c

Female slide 19

13-Which of the following lung tissues lack smooth muscle?

A. Alveoli

B. Trachea

C. Bronchus

D. Bronchioles

Ans: A

Female slide 20

14- Which cell type is primarily responsible for gas diffusion in the alveoli?

A. Type I pneumocytes
B. Type II pneumocytes
C. Alveolar macrophages
D. Club cells

Ans: A

Female slide 20

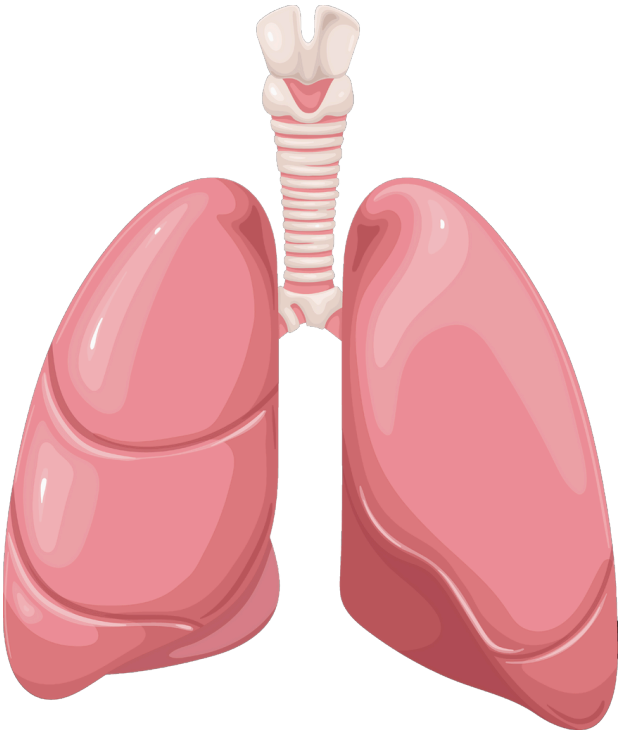
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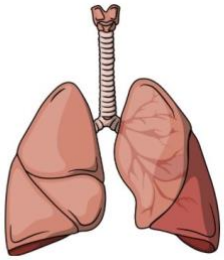
وقلوبنا بخشيتك،

وأسرارنا بطاعتك،

إنّك على كلّ شيء قدير،

وحسبنا اللَّه ونعم الوكيل.



ANATOMY 

L1 - Nasal cavity & paranasal sinuses

(16 Q's)

1- Olfactory mucosa is present in?

A. Roof of nasal wall

B. Floor of nasal wall

C. Over Inferior conchae

D. Lower part of medial wall

Ans: A

Female edit slide 10

2- All the following innervate the nasal cavity EXCEPT?

A. Mandibular

B. Maxillary

C. Superior ethmoidal

D. Olfactory

Ans: A

Female edit slide 10,13

3- What is the most labile parasinus to infections?

A. Sphenoidal

B. Maxillary

C. Frontal

D. Anterior ethmoidal

Ans: B

Female edit slide 31

4- Which of the paranasal sinuses is more liable to get infected?

A. Anterior ethmoidal

B. Frontal

C. Maxillary

D. Sphenoidal

Ans: C

Female edit slide 31

5- A 40-year-old male has an infection in the nasal cavity and also in the maxillary air sinus. This infection is making a mass in the hiatus and blocking it.

Where is the block located?

A. Superior meatus

B. Inferior meatus

C. Middle meatus

D. Sphenoethmoidal recess

Ans: C

Female edit slide 20

6- Which of the following bones is present in the nasal septum?

A. Frontal

B. Palatine

C. Vomer

D. Nasal

Ans: C

Female edit slide 14

7- Which of the following paranasal sinuses open in the sphenoethmoidal recess?

A. Sphenoid

B. Frontal

C. Maxillary

D. Posterior ethmoidal

Ans: A

Female edit slide 30

8- A 53-year-old patient has difficulty breathing through the nose. On examination, his physician finds that he has swelling of the mucous membrane of the superior nasal meatus. Which of the following paranasal sinuses is affected?

A. Middle ethmoidal

B. Posterior ethmoidal

C. Frontal

D. Maxillary

Ans: B

Female edit slide 20

9- A 53-year-old patient has difficulty breathing through his nose. On examination, the physician finds increased mucosa in the middle meatus. There is a bulging in which sinus?

A. Maxillary sinus

B. sphenoid sinus

C. Posterior ethmoidal sinus

D. Anterior ethmoidal sinus

Ans: A

Female edit slide 30

10- A young girl is complaining of a dry palate and nose. The lesion is most probably occurring in which ganglion?

A. Submandibular

B. Ciliary

C. Pterygopalatine

D. Otic

Ans: C

Female edit slide 22

11- Pointing to the roof of the nose, what is the special nerve?

A. Olfactory nerve
B. Optic nerve
C. Trigeminal nerve
D. Vagus nerve

Ans: A

Female edit slide 22

12- What sinus does this structure open to?

Marked structure: bulla

A. Middle ethmoidal sinus
B. Posterior ethmoidal sinus
C. Anterior ethmoidal sinus
D. Maxillary sinus

Ans: A

Female edit slide 20

13- A 53-year-old patient has difficulty breathing through his nose. On examination, his physician finds swelling of the mucous membrane of the superior nasal meatus. Which of the following paranasal sinuses is affected?

A. Middle ethmoidal air sinus
B. Posterior ethmoidal air sinus
C. Frontal air sinus
D. Maxillary air sinus

Ans: B

Female slide page 19

14- Which of the following bones contributes to the nasal septum?

A. Frontal bone

B. Palatine bone

C. Vomer bone

D. Nasal bone

Ans: C

Female slide page 14

15- A young girl is complaining of dryness of the palate and nose. The lesion is most likely located in which of the following ganglia?

A. Pterygopalatine ganglion

B. Ciliary ganglion

C. Submandibular ganglion

D. Otic ganglion

Ans: A

Female slide page 21

16- Which of the following air sinuses opens into the hiatus semilunaris?

A. Middle ethmoidal air sinus

B. Anterior ethmoidal air sinus

C. Frontal air sinus

D. Maxillary air sinus

Ans: D

Female slide page19

L2 - Anatomy of larynx

(20 Q's)

1- Contraction of cricothyroid muscle lead to?

A. Increased tension of vocal cords

B. Relaxation of vocal cords

C. Adduction of vocal cords

D. Abduction of vocal cords

Ans: A

Female slide 31, 34

2- Which of the following muscles causes adduction of vocal cord?

A. Posterior coricoarytenoid

B. Lateral coricoarytenoid

C. Cricothyroid

D. All of the above

Ans: B

Female slide 32

3- The laryngeal inlet is widened by?

A. Transverse arytenoid muscles

B. Oblique arytenoid muscles

C. Aryepiglottic muscles

D. Thyroepiglottic muscles

Ans: D

Female slide 29

4- The narrowest part of the larynx is?

A. Rima epiglottis

B. Laryngeal inlet

C. Rima vestibule

D. Infraglottic cavity

Ans: D

Female slide 3, 21

Rima glottidis is in the infraglottic cavity

5- During the surgery, if the surgeon accidentally cuts the accompanying nerve with the inferior thyroid artery, which of the following will be affected?

A. Paralysis of Vocalis Muscle

B. Paralysis of Cricothyroid Muscle

C. Decrease in Tension of Vocal Cords

D. Asphyxia and death

Ans: A

Male slide 73 , Edited Male slide 85

6- Which of the following muscles close the laryngeal inlet?

A. Thyroepiglottic muscle

B. Aryepiglottic muscle

C. Cricothyroid muscle

D. Thyroarytenoid muscle

Ans: B

Female slide 28, 29

7- A 53-year-old man underwent thyroid surgery and developed hoarseness in his voice due to nerve damage. Which of the following nerves most likely caused this damage?

A. External Laryngeal

B. Internal Laryngeal

C. Recurrent Laryngeal

D. Inferior Laryngeal

Ans: C

Female Edit slide 40, Male slide 73 , Edited Male slide 85

8- Which of these Nerves pierces the thyrohyoid membrane?

A. Superior Laryngeal
B. External Laryngeal
C. Internal Laryngeal
D. Recurrent Laryngeal

Ans: C

Female slide 12, 40

9- (Case) A patient with a benign thyroid tumor underwent thyroidectomy and then lost the ability to abduct his vocal cords. Which of the following muscles is affected?

A. Posterior Cricoarytenoid
B. Lateral Cricoarytenoid
C. Cricothyroid
D. Transverse Arytenoid

Ans: A

Female slide 27, 31, 32

10- Which muscle is responsible for widening the laryngeal inlet?

A. Transverse Arytenoid
B. Thyroepiglottic
C. Oblique Arytenoid

D. Aryepiglottic

Ans: B

Female slide 27,28,29

11- Which sensory nerve supplies the area above the vocal cords?

A. Internal laryngeal

B. Recurrent laryngeal

C. External laryngeal

D. Glossopharyngeal

Ans: A

Female slide 37

12- Which of the following nerves pierces thyrohyoid membrane?

A. Internal laryngeal

B. External laryngeal

C. Recurrent laryngeal

D. Superior laryngeal

Ans: A

Female slide 12

13- A surgery was performed on the superior thyroid artery, and it is suspected that nerve damage occurred. What would result from this nerve damage?

A. Decreased Tension

B. Increased Vocal Cord Length

C. Enhanced Pitch Control

D. Complete Vocal Cord Paralysis

Ans: A

Female Edit slide 40, Male slide 72 , Edited Male slide 84

14- A surgery was performed on the inferior thyroid artery, and it is suspected that nerve damage occurred. What would this nerve damage cause?

A. Loss of Sensation Above the Vocal Cords

B. Paralysis of Vocalis Muscle

C. Decrease in Tension of Vocal Cords

D. Inability to Produce High-Pitched Sounds

Ans: B

Female Edit slide 40, Male slide 73 , Edited Male slide 85

15- A patient is undergoing surgery to remove a goitre. The surgeon should be cautious to avoid damaging which of the following nerves?

A. Internal laryngeal nerve

B. External laryngeal nerve

C. Superior laryngeal nerve

D. Recurrent laryngeal nerve

Ans: D

Female Edit slide 40, Male slide 72, 73, 74, Edited Male slide 84,85

The recurrent laryngeal nerve is particularly at risk during thyroid surgery, as it innervates most of the intrinsic muscles of the larynx

16-Which nerve supplies the area above the vocal cords?

A. Internal laryngeal nerve
B. Recurrent laryngeal nerve
C. External laryngeal nerve
D. Glossopharyngeal nerve

Ans: A

Female slide 76

17- Which of the following nerves pierces the thyrohyoid membrane?

A. External laryngeal nerve
B. Recurrent laryngeal nerve
C. Internal laryngeal nerve
D. Glossopharyngeal nerve

Ans: C

Male slide 33

18- Which of the following muscles widens the laryngeal inlet?

A. Thyro-epiglottic muscle
B. Oblique arytenoid muscle
C. Aryepiglottic muscle
D. Cricothyroid muscle

Ans: A

Male slide 30

19- During a surgery involving the superior thyroid artery, nerve damage occurred. What would result from this injury?

A. Increased vocal cord length

B. Decreased tension of the vocal cords

C. Enhanced pitch control

D. Complete vocal cord paralysis

Ans: B

Male slide 35

20- A surgeon performed a thyroidectomy and accidentally injured a nerve that supplies this muscle. Which of the following muscles is affected?

A. Posterior cricoarytenoid

B. Lateral cricoarytenoid

C. Cricothyroid

D. Transverse arytenoid

Ans: C

Male slide 33

L3 - Anatomy of the pleura & lung

(21 Q's)

1- Left bronchus?

A. Wider than right

B. Vertical than right

C. It's bronchi divide after it's enter the lung

D. It's bronchi divide before it's enter the lung

Ans: C

Female slide 5

2- Horizontal fissure begins at?

A. 3rd costal cartilage

B. 4th costal cartilage

C. 5th costal cartilage

D. 6th costal cartilage

Ans: B

Female slide 23

3- Which lung lobe has 2 bronchopulmonary segment?

A. Middle lobe of right lung

B. Upper lobe of right lung

C. Middle lobe of left lung

D. Upper lobe of left lung

Ans: A

Female slide 29

4- In Bronchopulmonary segments, which of the following structures is intersegmental?

A. Segmental branch of pulmonary artery

B. Tributary pulmonary vein

C. Segmental bronchus

D. Lymph vessels

Ans: B

Female slide 35

5- The lower border of the plura cuts the mid-clavicular line at the level of? A. 6th

B. 7th

C. 8th

D. 9th

Ans: C

Female slide 16

6- Regarding the bronchopulmonary segment. Which of the following structures is considered intersegmental?

A. Segmental artery
B. Segmental vessel
C. Segmental vein
D. Segmental bronchus

Ans: C

Female slide 35

7- Which nerve supplies the diaphragmatic pleura?

A. Phrenic & intercostal
B. Intercostal
C. Vagus & phrenic
D. Recurrent laryngeal

Ans: A

Female slide 14

8- At which thoracic spine the oblique fissure cut at the posterior of lung?

A. Second
B. Third
C. Fourth
D. Fifth

Ans: B

Female slide 23

9- Which of the following lymph node drains the carnia of the trachea?

A. Tracheobronchial
B. Bronchopulmonary
C. Paratracheal
D. Pulmonry

Ans: A

Female slide 36

10- Which of the following lymph nodes drains the carina of the trachea?

A. Bronchopulmonary
B. Tracheobronchial
C. Hilar lymph nodes
D. Bronchopulmonary

Ans: B

Female slide 36

11- Why do aspirated bodies go into the right main bronchus ?

A. Beacuse the right bronchus is more vertical than the left bronchus
B. Beacuse the right bronchus is narrower than the left bronchus
C. Beacuse the right bronchus is more oblique than the left bronchus
D. Beacuse the right bronchus is longer than the left bronchus

Ans: A

Female slide 5

12- The lymph nodes present at the carina of the trachea are?

A. Bronchopulmonary
B. Tracheobronchial
C. Paratracheal
D. Pulmonary

Ans: B

Female slide 36

13- Regarding the bronchopulmonary segment, which of the following is intersegmental?

A. Bronchial artery
B. Segmental bronchi
C. Tributary pulmonary vein
D. Pulmonary artery

Ans: C

Female slide 35

14- The transverse fissure of the right lung ends anteriorly at the level of the costal cartilage number?

A. Fourth
B. Fifth

C. Sixth

D. Third

Ans: A

Female slide 23

15- What nerve supplies the mediastinal pleura?

A. Phrenic nerve

B. Intercostal nerve

C. C vagus nerve

D. D bronchial nerve

Ans: A

Female slide 14

16- An infant aged six months admitted to paediatric emergency , with dehydration due severe attack of gastroenteritis , after clinical examination, the physician described your urgent intravenous fluid only EJV was available to give , after few minutes the physician noted severe dyspnoea and respiratory distress of infant , X-ray showed pneumothorax of same side of injury, which part of pleura was suspected?

A. Costal

B. Visceral

C. Cervical

D. Mediastinal

Ans: C

Male edit slide 14

17- What is the nerve supply of the mediastinal pleura?

A. Sympathetic nerves
B. Phrenic nerve
C. Bronchial nerve
D. Intercostal nerves

Ans: B

Female edit slide 14

18- Horizontal fissure pierces the anterior border of the lung at the level of which costal cartilage?

A. Fifth costal cartilage
B. Sixth costal cartilage
C. Fourth costal cartilage
D. Third costal cartilage

Ans: C

Female edit slide 23

19- Which of the following lymph nodes are found inside the lung?

A. Pulmonary LNB
B. Bronchopulmonary LN
C. Tracheobronchial LN

D. Paratracheal LN

Ans: A

Female edit slide 37

20- Aspirated foreign bodies most commonly enter the right main bronchus due to which of the following reasons?

A. The right bronchus is narrower
B. The right bronchus is more oblique
C. The right bronchus is longer
D. The right bronchus is wider, shorter, & more vertical

Ans: D

Female edit slide 5

21- (Lung picture, arrow pointing to hilar part of the right lung) Which of the following arteries supplies it?

A. Right pulmonary artery
B. Right 3rd posterior intercostal artery
C. Left 3rd intercostal artery
D. Descending thoracic aorta

Ans: B

Female edit slide 26

L4 - Medial relations of the lungs

(7 Q's)

1- Which one of the following is related to the medial side of LT lung?

A. Esophagus

B. IVC

C. Azygos arch

D. LT atrium

Ans: A

Female slide 7

2- The trachea is in medial relations with?

A. Anterior superior lobe of left lung

B. Posterior inferior lobe of right lung

C. Posterior inferior lobe o left lung

D. Posterior superior lobe of right lung

Ans: D

Female slide 6

3- Which of the following is related to the medial side of the left lung?

A. Superior Vena Cava

B. Azygos arch

C. Descending aorta

D. Inferior Vena Cava

Ans: C

Female slide 7

4- Which of the following structures is related to the medial side of the left lung ?

A. Left atrium

B. Arch of aorta

C. SVC

D. Azygos

Ans: B

Female slide 7

5- Which of the following is related to the medial surface of the right lung ?

A. Right ventricle

B. Left atrium

C. Arch of azygous vein

D. Descending aorta

Ans: C

Female slide 6

6- What is related to the lower medial surface of the left lung?

A. Azygos arch
B. Left atrium
C. Esophagus
D. Trachea

Ans: C

Female slide 7

7- What is found on the medial side of the left lung?

A. Azygos vein
B. Arch of the aorta
C. Left atrium
D. SVC

Ans: B

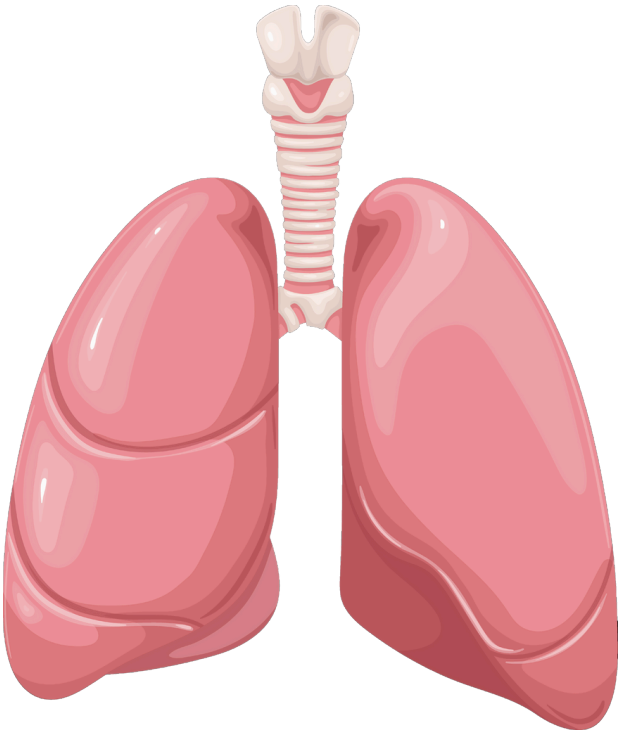
Female slide 7

8- Which of the following structures forms an impression on the lower medial side of the left lung?

A. Esophagus
B. Superior vena cava
C. Azygos arch
D. Left atrium

Ans: A ,Female edit slide7

اللهم ارزقني قوة الحفظ، وسرعة الفهم، وصفاء الذهن



EMBRYOLOGY

L1- Development of respiratory system

(13 Q's)

1- Which of the following congenital anomalies causes diaphragm hernia?

A. Tracheo-esophageal fistula

B. Extra lung lobe

C. Lung cyst

D. Hypoplasia of the lung

Ans: D

Female slide25

2- During lung development, repeated division of bronchi and alveoli continues till the age of?

A. 4 Years

B. 5 Years

C. 8 Years

D. 1 Year

Ans: C

Female slide 22

3- During maturation of the lung, the pseudoglandular stage has?

A. Terminal bronchioles
B. Respiratory bronchioles
C. Alveolar sac
D. Alveolar duct

Ans: A

Female slide 14

4- Surface area of alveoli continues to increase until?

A. Childhood
B. Adulthood
C. 8 Years
D. Before birth

Ans: B

Female slide 22

5- Which of the following is the most common congenital anomaly of the trachea and esophagus?

A. Esophageal atresia and tracheoesophageal fistula
B. H-type tracheoesophageal fistula
C. Esophageal atresia
D. Isolated esophageal atresia without fistula

Ans: A

Female slide 23

6- At the pseudoglandular stage, which of the following is present?

A. Alveolar ducts
B. Alveoli
C. Respiratory bronchioles
D. Terminal bronchioles

Ans: D

Female slide 14

7- During lung growth, at which stage does the surface area of alveoli increases?

A. Adult stage
B. Before birth
C. 8 Years
D. Childhood

Ans: A

Female slide 22

8- Regarding the embryonic origin of the respiratory system, the epithelial lining originates from?

A. Splanchnic layer
B. Mesoderm
C. Ectoderm

D. Endoderm

Ans: D

Female slide 6

9- The most common congenital anomaly in the development of the lung is?

A. Esophageal atresia

B. Hypoplasia

C. Agenesis

D. Tracheo-esophageal fistula

Ans: D

Female slide 23

10- The epithelium of the respiratory tract comes from which of the following?

A. Endoderm

B. Ectoderm

C. Splanchnic mesoderm

D. Somatic mesoderm

Ans: A

Female slide 6

11- Which of the following diseases is caused by incomplete separation of tracheoesophageal septa ?

A. Esophageal atresia

B. Trachea atresia

C. Tracheoesophageal fistula

D. Agenesis

Ans: C

Female slide 23

12- Which of the following structures is seen in pseudoglandular phase?

A. Terminal bronchioles

B. Respiratory bronchioles

C. Alveolar ducts

D. Alveolar sacs

Ans: A

Female slide 14

13- During maturation of the lung the pseudoglandular stage contains?

A. Respiratory bronchioles

B. Terminal bronchioles

C. Alveolar sacs

D. Alveolar ducts

Ans: B

Female slide 14

14- The epithelium of the respiratory tract comes from which of the following?

A. Endoderm
B. Ectoderm
C. Splanchnic mesoderm
D. Somatic mesoderm

Ans: A

Female slide 6

15- During maturation of the lung, the Canalicular stage contains which of the following?

A. Alveolar ducts
B. Alveolar sacs
C. Alveoli
D. Primitive alveoli

Ans: A

Female slide 15

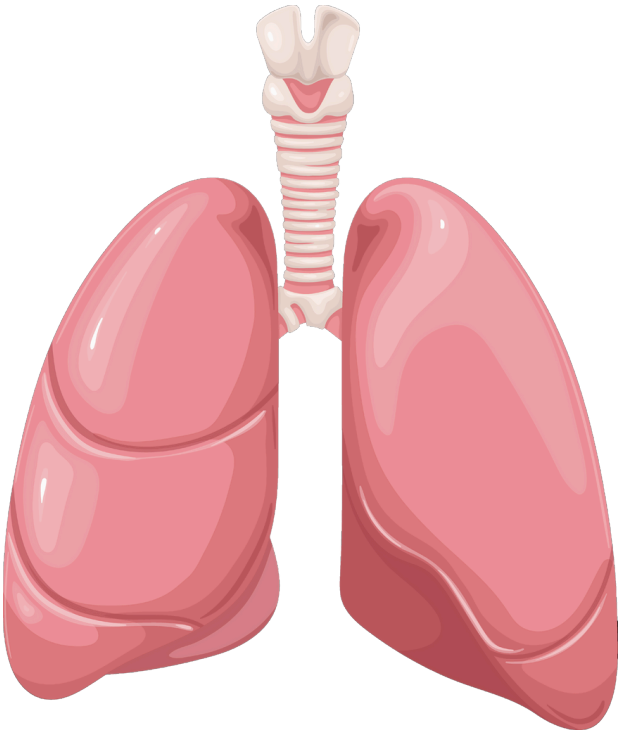
16- Surface area of alveoli continues to increase until:

A. Adult stage
B. Before birth
C. 8 years
D. Childhood

Ans: A

Female slide 22

اللهم افتح لي ابواب حكمتك،
وانشر علي رحمتك،
وآمن علي بالحفظ والفهم،
سبحانك لا علم لنا الا ماعلمتنا،
انك انت العليم الحكيم .



MICROBIOLOGY



L1- Upper respiratory tract infection

(11 Q's)

1- What is the virulence factor of Corynebacterium diphtheriae?

A. Enterotoxin

B. Exotoxin

C. Enzyme

D. Capsule

Ans: B

Female slide 27

2- (Case) A patient came with fever, pharyngeal edema, bull neck appearance, and gray membrane coating the tonsils. What is the organism?

A. Corynebacterium diphtheriae

B. Streptococcal tonsillitis

C. Staphylococcal aureus

D. Streptococcus pneumonia

Ans: A

Female slide 27

3- (Case) A patient came with inflammation of the pharynx, tonsils, and gums and offensive halitosis caused by borrelia vincentii and fusobacterium. What is the diagnosis?

A. Vincent's Angina
B. Epiglottitis
C. Croup
D. Viral pharyngitis

Ans: A

Female slide 25

4- (Case) A patient came with high fever, purulent exudates, and enlarged cervical lymph node. What is the immunological complication of this disease?

A. Rheumatic fever
B. Scarlet fever
C. Septicemia
D. Glandular fever

Ans: A

Female slide 16

5- A patient comes to the clinic and has inflammation in the pharynx, tonsils, and gums with necrotic debris. Gram stains reveal a gram-negative spiral organism and gram-negative long tapering bacilli. What is the disease?

A. Streptococcal tonsillitis
B. Epiglottitis
C. Fever
D. Vincent angina

Ans: D

Female slide 25

6- The disease that can be prevented with vaccination is?

A. Epiglottitis
B. Legionellae's disease
C. Croup
D. Common cold

Ans: A

Female slide 23

7- The following microorganism is the most cause of acute otitis externa?

A. Haemophilis Influenza
B. Pseudomonas aeroigenosa
C. Streptococcus pneumonia

D. B-haemolytic streptococci

Ans: B

Female slide 33

8- What is the mode of transmission of rhinovirus ?

A. Aerosol

B. Food

C. Contact

D. Blood

Ans: A

Female slide 5

9- How is croup transmitted?

A. Blood

B. Aerosol

C. Contact

D. Food

Ans: B

Female slide 13

10- What is the treatment of acute epiglottitis?

A. Ampicillin

B. Amantadine

C. Cefotaxime

D. Penicillin

Ans: C

Female slide 23

11-(case) 3 year old baby has bovine cough the most likely organism?

A- RSV

B- Para influenza viruse

C- Paravirus

D-Rhinovirus

Ans:B

Edited female slide 15

L2- Lower respiratory tract infection

(20 Q's)

1- **(Case) A child comes to the hospital with paroxysmal cough, cyanosis, eye congestion, high pitched sound during intaking air between bouts of coughing.**

He was infected with gram negative coccobacilli. What is the diagnosis?

A. Whooping cough

B. Epiglottitis

C. Severe acute respiratory syndrome (SARS)

D. Tuberculosis

Ans: A

Female slide 29

2- (Case) 5 months infant with bronchiolitis ?

A. Respiratory syncytial virus

B. Adenovirus

C. Influenza virus

D. Rhinovirus

Ans: A

Female slide 22

3- A 44 year old patient came to you complaining of a persistent cough for three weeks, night sweats, fever and weight loss. Ziel Neelsen stain showed red beaded bacilli in a blue background. what is the virulence factor of its pathogenesis?

A. Bacteria can destroy macrophages

B. Bacteria can survive in macrophages

C. Bacteria can live in bronchi

D. Bacteria can destroy pulmonary tissue

Ans: B

Female slide 32

4- A patient presented with lower respiratory tract symptoms, he give history of bird keeping, what is the causative organism?

A. Bordetella

B. Diphtheria

C. SARS

D. Chlamydia psittaci

Ans: D

Female slide 11

5- What is the most common cause of acute bronchitis?

A. Bacterial infection

B. Viral infection

C. Fungul infection

D. Non-infectious cause

Ans: B

Female slide 14

6- A 68 year old diabetic man presented to the ER with productive cough, purulent sputum, and shortness of breath. Chest X-ray shows consolidation

in the lobe of the lung. What organism is most likely the cause of his symptoms?

A. Chlamydia Psittacosis

B. Haemophilus Influenzae

C. Streptococcus Pneumonia

D. Staphylococcus Aureus

Ans: C

Female slide 5

7- A 60 years old male goes to Alulaa , after 2-3 days he start to complain of runny nose, cough, sore throat, and fever , also its progressed to acute pneumonia and diarrhea . What is the organism that he affected with after contact with camels ?

A. SARS

B. MERS-COV

C. Chlamydia pneumonia

D. Staphylococcus aureus

Ans: B

Female slide 17

8- Which of the following is the most causative organism to community acquired pneumonia ?

A. Streptococcal pneumonia

B. Mycoplasma pneumonia

C. Respiratory syncytial virus

D. Candida

Ans: A

Female slide 5

9- Species of the following genus causes zoonotic disease?

A. Legionella

B. Chlamydia

C. Bordetella

D. Escherichia

Ans: B

Female slide 11

10- A ten month old boy presented to the clinic with cough, fever, runny nose, and otitis media. Pulmonary examination showed him to have bronchiolitis. These clinical findings are most commonly caused by?

A. Rhinovirus

B. Respiratory syncytial virus

C. Para influenzas

D. Adenovirus

Ans: B

Female slide 22

11- BCG vaccine is given?

A. Intradermal
B. Intramuscular
C. Subcutaneous
D. Oral

Ans: A

Female slide 36

12- A 30 years old female, under chemotherapy, was diagnosed with an infection. What's the most common microorganism that affects immunocompromised patients?

A. Salmonella
B. Leishmanial
C. Aspergillosis
D. Pneumonia jiroveci

Ans: C

Female slide 26

13- Which of the following is not taken into consideration when prescribing a drug for community acquired pneumonia?

A. Family history of lung disease

B. Previous antibiotic use

C. Excessive alcohol intake

D. Co-morbidities

Ans: A

Female slide 27

14- (Case) A patient presents with symptoms of dyspnea, cough, and malaise. She has not traveled recently. Laboratory tests indicate leukocytosis with an abundance of neutrophils and hemoptysis. A chest X-ray reveals consolidation in the right middle lobe.

A. Primary tuberculosis
B. Pneumonia
C. Bronchitis
D. Asthma

Answer: B

Female slide 3

15- Case about an HIV patient; on microscope they found cup-shaped cysts in silver stain, most common likely organism?

A. Histoplasma
B. Coccidioides
C. Pneumocystis jiroveci
D. Candidiasis

Answer: C

Female slide 26

16- A 74 y/o male presented to the emergency with lethargy , fever , productive cough , worsening dyspnea for a week duration . Diagnosis was demonstrate severe pneumonia. Which of the following is complication of bacterial pneumonia?

A. Lung cancer
B. Heart failure
C. Lung fibrosis
D. Empyaema

Answer: D

Female slide 3

17- 45-year-old male with a past history of smoking and complains of fever, mild diarrhea, nonproductive cough. Gram staining of sputum shows no microorganisms. A urine test was performed and a bacterial antigen was found. The MOST recommended treatment for this infection is expected to be:

A-Macrolide

B-Metronidazole

C-Tetracycline

D-ceftriaxone

Ans:A. Female slide9

18-Which of the following microorganisms causes Q fever?

A-Streptococcus pyogenes

B-Coxiella burnetii

C-Streptococcus viridans

D-Legionella pneumophila

Ans:B

Female slide 14

19-Which of the following is a complication of streptococcal pneumonia?

A-Pneumothorax

B-Pulmonary Embolism

C-Tuberculosis

D-Para-pneumonic effusion

Ans:D

Female slide 6

20-A patient presents with fever and features of lower respiratory tract infection. He gives the history of keeping birds and parrots. What is the causative organism?

A- Mycoplasma pneumoniae

B- Legionella pneumophila

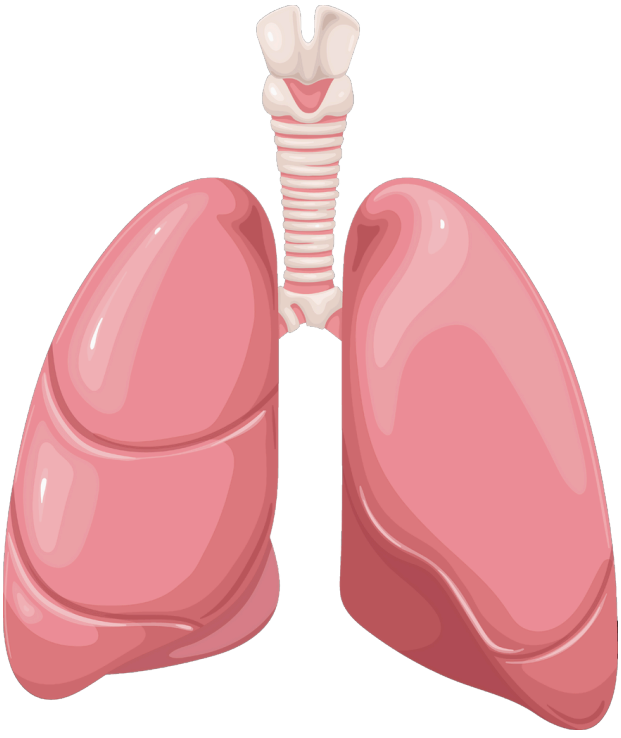
C- Chlamydia psittaci

D- Respiratory syncytial virus (RSV)

Ans:C

Female slide 13

اللهم يا جامع النّاس في يوم لا ريب فيه
اجمع عليّ ضالتي



PARASITOLOGY

L1- Parasitic cysts of the lungs

(15 Q's)

1- The causes of anaphylactic shock?

A. Entamoeba histolytica

B. Hydatid cyst

C. Trematodes

D. Ascaris lumbricoides

Ans: B

Female slide 34

2- Infection with hydatid cyst occurs after?

A. Ingestion of parasitic cyst

B. Eating an organ with hydatid cyst

C. Ingestion of Echinococcosis granulosus egg

D. Inhalation of contaminated water

Ans: C

Female slide 27

3- A 37 year old Patient came with dyspnea,chest pain,and cough. ct scan and chest x-ray showed that there is pleural effusion and right lung abscess.what is the most causative agent?

A. Ascaris
B. Echinococcus garnulus
C. Entamoeba histolytica
D. Streptococcus pneumoniae

Ans: C

Female slide 15

4- A Patient go trip in alula and after two weeks he have upper right quadrant pain , jaundice, and abdominal mass with lung abscess. During surgery they find liver cyst. What the parasite that he infected with?

A. Echinococcus granulosus
B. Schistosomae
C. Entamoeba histolytica
D. Paragonimus westermani

Ans: A

Female slide 33

5- Larval form of Echinococcus granulosus is present in which host?

A. Dog
B. Wolf
C. Man

D. Sheep

Ans: C

Female slide 23

6- A patient presented with cough, dyspnea, chest pain. CXR/investigation revealed a small amount of pleural effusion and abscess in the lung. What is the most likely causative parasite?

A. Entamoeba histolytica
B. Echinococcus granulosus
C. Paragonimus westermani
D. Schistosomiasis

Ans: A

Female slide 15

7- A patient came to the ER with a cough and fever, X-rays showed elevated right diaphragm and lower right pleural effusion and abscess. What is the causative agent?

A. Entamoeba histolytica
B. Ascariasis
C. Hydatid cyst
D. Toxocariasis

Ans: A

Female slide 16

8- A patient came in with skin flushed, they found a ruptured hepatic cyst. What is the causative agent?

A. Echinococcus
B. Ascariasis
C. Toxocariasis
D. Entamoeba histolytic**a**

Ans: A

Female slide 33

9- What is the preventative measure for Echinococcus?

A. Removal of snails
B. Avoid contamination of hands, food, and water with dog feces
C. Wear preventive clothes (gloves - shoes)
D. Proper washing of raw vegetables

Ans: B

Female slide 38

10- A male shepherd from AlUla came complaining of hepatomegaly, upper right quadrant pain, jaundice, and a non-painful hepatic cyst was found. What is the likely causative agent?

A. Echinococcus
B. Entamoeba histolytica
C. Schistosomiasis

D. Ascaris lumbricoides

Ans: A

Female slide 33

11- Where are the larval forms of Echinococcus found?

A. Man

B. Dog

C. Cat

D. Wolf

Ans: A

Female slide 23

12- What is the common extrahepatic site of ambiosis?

A. Peritoneum

B. kidney

C. Spleen

D. Lung

Ans: D

Female 15

13- A patient comes to the clinic with a chronic cough, fever, and right-sided chest pain. Imaging shows an elevated right diaphragm, What is a likely parasitic cause?

A. Entamoeba histolytica
B. Schistosomiasis
C. Toxocariasis
D. Ascaris lumbricoides

Ans: A

Female slide 16

14- What's the infective stage of entamoeba histolytica?

A. Egg
B. Trophozoites
C. Cyst
D. Cercaria

Ans: C

Female slide 13

15- A 57 year old farmer presents with dyspnea, chest pain, and a chronic cough. CXR shows homogeneous opacities that almost fill up the right lung. Which of the following is the most likely diagnosis?

A. Amoebiasis
B. Ancylostomiasis
C. Toxocariasis
D. Echinococcosis

Ans: D

Female slide 34

L2- Parasites causing visceral larva migrates

(9 Q's)

1- Loffler's Syndrome in the lung is due to?

A. Sensitization to ascaris antigen

B. inflammation reaction against ascaris larva in the lung

C. viral infection in the lung

D. Bacterial infection in the lung

Ans: B

Female slide 13

2- Which of the following is a prevention of Ancylostoma?

A. Wear shoes

B. Wash vegetables

C. Wash hands

D. Proper sewage disposal

Ans: A

Female slide 25

3- Which one of the following infections lead to Cor pulmonale?

A. Toxocariasis
B. Hydatid
C. Amoebiasis
D. Schistosomiasis

Ans: D

Female slide 42

4- A 42 year old man returned two weeks ago from a five month trip to Ghana. During his stay he had frequent visits to Lake which included swimming in the lake. He was present with fever, cough wheeze myalgia headache and urticaria.stool analysis revealed 2-3 eggs of schistosoma mansoni. What is the stage?

A. chronic schistosomiasis
B. Cercarial penetration (invasion stage)
C. Cercaria (migration stage)
D. Acute Schistosomiasis (Katayama)

Ans: D

Female slide 39&40

5- Which of the following is the egg that causes Ascaris infection to humans by raw vegetables and contaminated water?

A. Fertilised egg

B. Unfertilized egg

C. Mature egg

D. Immature egg

Ans: C

Female slide 10

6- Patient presented with 3 weeks of cough, dyspnea, CXR revealed bilateral zone patch consolidation, eosinophilia and Ascaris larva in sputum, what is this patient have ?

A. Katameya fever

B. Liver abscess

C. Lung cyst

D. Loffler's syndrome

Ans: D

Female slide 14

7- Which of the following can lead to Cor pulmonale?

A. Schistosomiasis

B. Toxocariasis

C. Echinococcosis

D. Ascariasis

Ans: A

Female slide 42

8- A 6-year-old boy played with dogs on a school trip, later he came in with chest tightness,and bronchospasm. What is the possible organism?

A. Toxocara
B. Schistosoma
C. Ascaris
D. Ancylostoma

Ans: A

Female slide 26&30

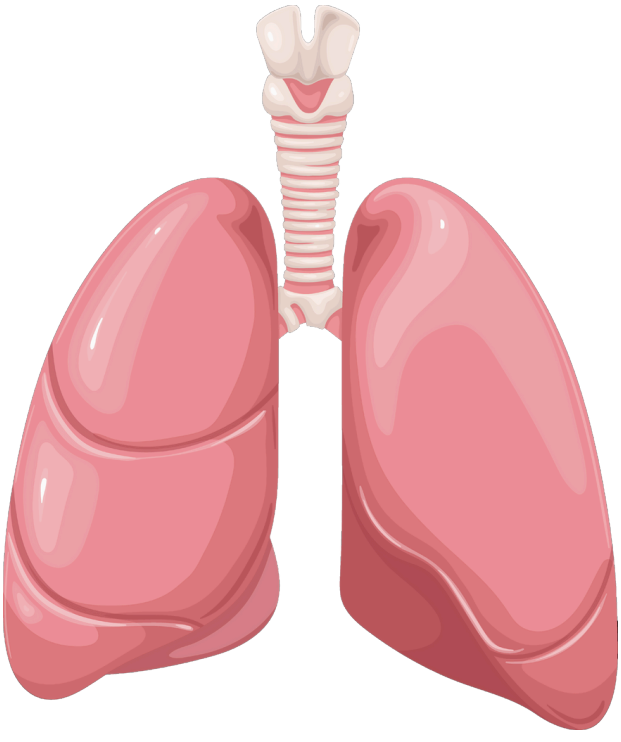
9- Patient ate green salad and got a parasitic infection with increased eosinophilia?

A. Ascariasis
B. Schistosomes
C. Toxocariasis
D. Entamoeba histolytica

Ans: A

Female slide 10&15

اللهم افتح لي ابواب حكمتك،
وانشر علي رحمتك،
وآمن علي بالحفظ والفهم،
سبحانك لا علم لنا الا ماعلمتنا،
انك انت العليم الحكيم .



BIOCHEMISTRY 

L1- Metabolic functions of the lungs & pulmonary surfactant

(13 Q's)

1- During surfactant synthesis, which of the following is found extra-cellular in alveolar lumen and repair SP-A & SP-B for its formation and stabilization?

A. DPPC

B. Type II alveolar cells

C. Lamellar body

D. Tubular myelin

Ans: D

Female slide 39

2- During the lung maturity test a pregnant woman's L/S ratio is determined to be 1:1, what can we conclude from this result?

A. Decreased risk of respiratory distress syndrome

B. No specific significance is concluded

C. Insufficient production of surfactant

D. Adequate lung maturity

Ans: C

Female slide 47

3- A baby is born at 37 weeks with normal pulmonary function. A pulmonary test was performed; what is the likely finding?

A. Phosphatidylglycerol is going to make up 90% of surfactant
B. Presence of Phosphatidylglycerol
C. L/S ratio is below 2
D. Sphingomyelin is the prominent surfactant

Ans: B

Female slide 49

4- A baby born at 37 weeks has a good respiratory function, which of the following can indicate that ?

A. L/S ratio < 2
B. Surfactant contains phosphatidylglycerol
C. Surfactant contains sphingomyelin
D. L\S ratio = 1

Ans: B

Female slide 49

5- A full-term newborn was brought to the hospital with pale skin and peripheral cyanosis. A genetic analysis was performed and revealed a

hydrophobic protein deficiency causing respiratory distress. What is the missing protein?

A. SP-B

B. SP-D

C. Surfactant

D. Dipalmitoyl phosphatidyl choline

Ans: A

Female slide 36, Female edit 37

6- Which surfactant protein is responsible for the synthesis of tubular myelin and its deficiency causes infant disease?

A. SP B

B. SP D

C. SP A

D. SP C

Ans: A

Female slide 36, Female edit 37

7- A 32 year old woman came to assess her L/S ratio to assess fetal lung maturity. Her L/S ratio was less than 1. What was the expected gestational age?

A. 38 weeks

B. 35 weeks

C. 32 weeks

D. 30 weeks

Ans: D

Female slide 40, Female edit 41

8- A sensitive lung maturity test that includes automated counting of phospholipid granules from the amniotic fluid is?

A. L/S

B. Lamellar body count

C. Phosphatidylglycerol

D. D.Surfactant/albumin ratio

Ans: B

Female slide 51

9- During the synthesis of surfactant, it is secreted into the alveolar lumen and needs SP-A and SP-B for its synthesis and stabilization?

A. Lamellar bodies

B. Type II pneumocyte

C. Tubular myelin

D. DPPC

Ans: C

Female slide 39

10- The main component of surfactant?

A. dipalmitoyl phosphatidyl choline
B. Phosphatidylinositol
C. Phosphatidylglycerol
D. Protein

Ans: A

Female slide 27, Female edit 28

11- A newborn baby was born 9 weeks prematurely. Her aspiration is low, and her lips and fingers are cyanotic (blue) from low oxygen levels in her arterial blood. Her condition is related to a problem with the synthesis of lipoprotein.

What is the main constituent of this lipoprotein?

A. Dipalmitoyl phosphatidyl chloride
B. Phosphatidic acid
C. Angiotensin-converting enzyme
D. Cholesterol

Ans: A

Female slide 28

12- The lung has an ability to protect its systemic vasculature from vasoactive substances by?

A. Release of histamine
B. Synthesis of surfactant
C. Release of bradykinin and norepinephrine
D. Synthesis and release of norepinephrine

Ans: A

Female slide

13- Pulmonary surfactant component:

A. Protein and phospholipid
B. Protein and carbohydrate
C. Nucleic acid and protein
D. Fatty acid and enzyme

Ans: A

Female slides page 25

L2- Acid-base balance

(17 Q's)

1- Which of the following is the main source of CO_2 production?

A. Incomplete oxidation of CHO and Fats
B. Oxidative metabolism
C. Oxidation of sulfa-containing amino acids
D. Anaerobic respiration of glucose

Ans: B

Female med22 slide 12

2- (Case) PH: 7.48, CO_2: Low?

A. Asthmatic patient
B. Anxiety hyperventilation
C. Severe diarrhea
D. Vomiting

Ans: B

Female med22 slide 47

3- Which of the following is found in high concentration and is therefore the most effective buffer system at physiological pH?

A. Phosphate buffer
B. Bicarbonate buffer
C. Hemoglobin
D. Plasma protein buffer

Ans: B

Female med22 slide 32

4- Which of the following factors shift the oxyhemoglobin curve to the left?

A. Acidosis

B. Hyperthermia

C. Exercise

D. Hyperventilation

Ans: D

Female med22 slide 17

5- During cell metabolism, the cells produce CO2, which then combines with water to form carbonic acid. What is the role of hemoglobin in this situation?

A. Clear out excess acidosis by taking H in the imidazole ring

B. Clear excess alkalosis by releasing H

C. Clear out excess acidosis with the help of leucine

D. Clear excess acidosis by releasing H from its carboxyl terminal

Ans: A

Female med22 slide 22

6- A patient with hypoventilation is most likely to have which of the following?

A. High plasma bicarbonate

B. Hypocapnea

C. Cool extremities

D. Hypercapnea

Ans: D

Female med22 slide 43

7- Which of the following is the main source of CO2 production?

A. Incomplete oxidation of CHO and Fats

B. Oxidative metabolism

C. Oxidation of sulfa-containing amino acids

D. Anaerobic respiration of glucose

Ans: B

Female med22 slide 12

8- Which of the following is the immediate effect of high-altitude climbing?

A. Hyperventilation

B. Polycythemia

C. Acidosis

D. Increased 2,3-DPG

Ans: A

Female med22 slide 54

9- What is the mechanism of the hemoglobin buffer system?

A. Chloride shift
B. Neutralize excess acidity via uptake the H by imidazole histidine ring
C. Neutralize excess alkalinity via uptake the H by imidazole histidine ring D. Shift the curve to the right

Ans: B

Female med22 slide 27

10- Why is venous blood more acidic than arterial blood?

A. Deoxygenated hemoglobin has more affinity for H+
B. No carbonmyl enzyme
C. Oxygenated hemoglobin has more affinity for H+
D. Less oxygen

Ans: A

Female med22 slide 29

11- A 35-year-old diabetic female presents with abdominal pain and vomiting after attending a party. Her blood sugar is 600 mg/dl (normal <140 mg/dl). The values obtained from the ABG were: pH=7.03 (normal 7.35-7.45), PCO2=36 mm Hg (normal 35-45 mm Hg), HCO3-=14 mEq/L (normal 22-26 mEq/L). How will the body compensate?

A. Increase H+ to buffer HCO3-
B. Hypoventilate
C. Decrease HCO3- reabsorption

D. Hyperventilate

Ans: D

Female med22 slide 68

12- Which amino acid in hemoglobin helps buffer blood through imidazole ring?

A. Histidine

B. Glycine

C. Alanine

D. Serine

Ans: A

Female slides page 26,27

13- A patient with pH 7.22, PCO_2 55, and normal HCO_3^-. What is the diagnosis?

A. Respiratory acidosis

B. Respiratory alkalosis

C. Metabolic acidosis

D. Metabolic alkalosis

Ans: A

Female slides page 49

14- What is the source of volatile H^+?

A. Lactic acid

B. Phosphoric acid

C. Carbonic acid

D. Sulfuric acid

Ans: C

Female slides page 11

16- Which of the following metabolic disorders is caused by hypoventilation?

A. Respiratory acidosis

B. Respiratory alkalosis

C. Metabolic acidosis

D. Metabolic alkalosis

Ans: A

Female slides page 44

17- Which of the following factors shift the oxyhemoglobin curve to the right?

A. Decreased CO_2 pressure

B. Decreased DPG

C. Increased pH (alkalosis)

D. Increased temperature

Ans: D

Female slides page 18

L3- Arterial blood gases

(10 Q's)

1- The sum of O2?

A. Oxygen content

B. PaO2

C. SpO2

D. SaO2

Ans: A

Female edit slide 17

2- Which of the following shows the amount of Oxygen in the blood?

A. Oxygen content (Ca O2)

B. Oxygen Saturation

C. PO2

D. None

Ans: A

Female edit slide 17

3- Limitations of ABG testing?

A. Not Detecting acid-base imbalance

B. Not Detecting bicarbonate levels

C. Giving diagnosis of the disease

D. It can be used as a point of care testing

Ans: C

Female edit slide 37

4- Which of the following best reflects the sum of the O2 bound to hemoglobin and that dissolved in plasma?

A. Partial O2 pressure

B. Oxygen dissociation curve

C. O2 content

D. O2 saturation

Ans: C

Female edit slide 17

5- When comparing SaO_2 with PaO_2, which of the following is correct? A. If PaO_2 is 80-100 mmHg, it is normal; if SaO_2 is >95%, it is normal.

B. SaO_2 reflects oxygen in plasma.

C. SaO_2 can be affected by carbon monoxide only.

D. PaO_2 is unaffected by hemoglobin levels.

Ans: A

Female edit slide 11&12

6- Which of the following best reflects percent oxygenation hemoglobin?

A. Oxygen content

B. Arterial pressure

C. Oxygen saturation

D. PaO_2

Ans: C

Female edit slide 12

7- Summary of a long question: pH=7.34, PCO_2 = 70, HCO_3/H_2CO_3 = 20:1, what does this state represent?

A. Compensatory Respiratory Acidosis

B. Uncompensatory Respiratory Acidosis

C. Compensatory Respiratory Alkalosis

D. Uncompensatory Respiratory Alkalosis

Ans: A

Female edit slide 35

8- A 20-year-old medical student came to the ER with his friend who described that during a test, he started sweating, breathing heavily, and felt dizzy. In the ER, his ABG showed pH 7.49, pCO2 (29 mmHg), and HCO3/H2CO3 ratio 20:1. What is the acid-base state of this patient?

A. Compensated respiratory alkalosis

B. Non-compensated respiratory acidosis

C. Compensated respiratory acidosis

D. Non-compensated respiratory alkalosis

Ans: A

Female edit slide 37

9- You are given an ABG report which states: PaO_2 = 80 mmHg, SaO_2 = 95%, and CaO_2 = 20 ml O_2/dl. Which of the following fits the description?

A. The total amount of oxygen in the blood is 80 mmHg

B. The oxygen bound to hemoglobin and that dissolved in the plasma is 20 ml O_2/dl

C. The partial pressure of oxygen in arterial blood is 20 ml O_2/dl

D. The oxygen content of venous blood is 80 mmHg

Ans: B

Female slides page 17

10- Client Z is admitted to the hospital and is to undergo brain surgery. The client is very anxious and scared of the upcoming surgery. He begins to hyperventilate and becomes very dizzy. The client loses consciousness and the STAT ABGs reveal pH 7.61, $PaCO_2$22 mmHg, and HCO_3^- 25 mEq/L. What is the ABG interpretation based on the findings?

A. Metabolic acidosis, uncompensated

B. Respiratory alkalosis, partially compensated

C. Metabolic alkalosis, partially compensated

D. Respiratory alkalosis, uncompensated

Ans: D

Female slides page 33